Current

Exam code: 0625 & 0972

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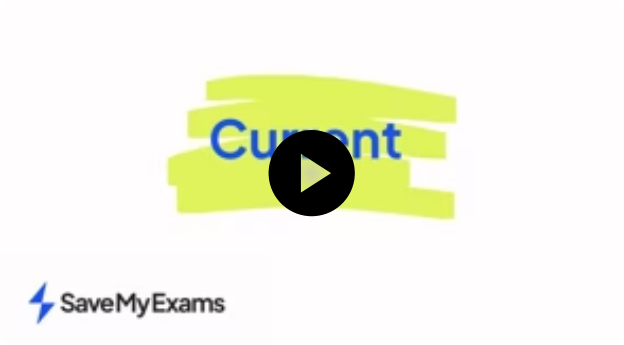

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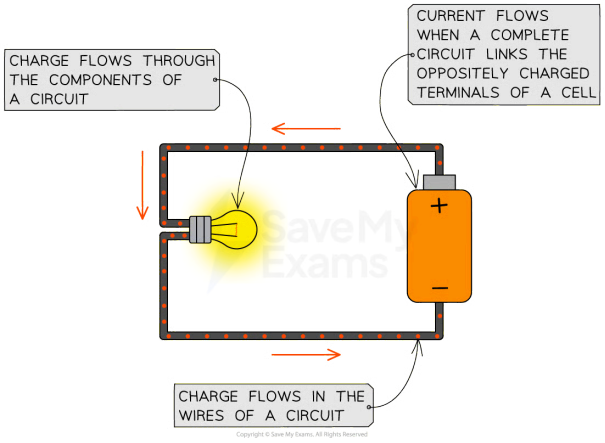
Current

- Electric current is defined as

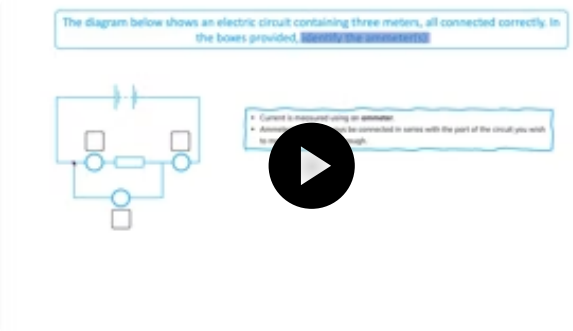
The rate of flow of electric charge

- Current flows
  - when a **circuit** is formed e.g. when a wire connects the two oppositely charged terminals of a cell
  - from the **positive** terminal to the **negative** terminal of a cell



Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



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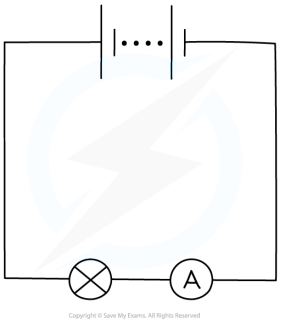
Measuring current

- Current can be measured using an **ammeter**
- Ammeters must be connected in **series** with the component being measured

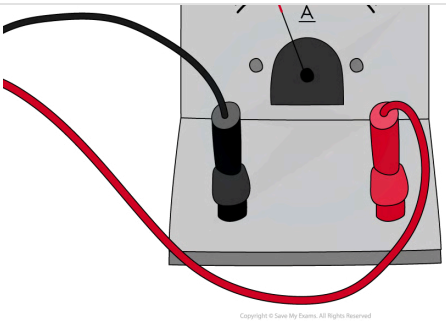


An ammeter can be used to measure the current around a circuit

- Ammeters can be
  - digital (with an electronic read out)
  - analogue (with a needle and scale)

Analogue ammeters

- Typical ranges are 0.1–1.0 A and 1.0–5.0 A for analogue ammeters
  - Always double check exactly where the marker is before an experiment, if not at zero, you will need to subtract this from all your measurements. They should be checked for **zero errors** before using
- They are also subject to **parallax error**
  - Always read the meter from a position directly perpendicular to the scale



Digital ammeters

- Digital ammeters can measure very small currents, in mA or μA
- Digital displays show the measured values as digits and are more accurate than analogue displays
- They're easy to use because they give a specific value and are capable of displaying more precise values
  - However digital displays may 'flicker' back and forth between values and a judgement must be made as to which to write down

- Digital ammeters should be checked for **zero error**
  - Make sure the reading is zero before starting an experiment, or subtract the "zero" value from the end results

Electrical conduction in metals

- The wires in an electric circuit are made of **metal** because it is a good **conductor** of electric **current**
- In the wires, the current is a flow of **negatively charged electrons**

Calculating current

Extended tier only

- Electric current can be defined more precisely as:

  The charge passing a point in a circuit per unit time

- Current is measured in units of **amperes** or **amps (A)**
  - 1 amp is equivalent to a charge of 1 coulomb flowing in 1 second, or 1 A = 1 C/s

- This means the size of an electric current is the amount of charge passing through a component each second
- Current, charge and time are related by the equation:

$$I = \frac{Q}{t}$$

- Where:
  - Q = charge, measured in coulombs (C)
  - I = current, measured in amps (A)
  - t = time, measured in seconds (s)

- The current, charge and time equation can be rearranged with the help of the following formula triangle:

Current charge time formula triangle

Formula triangle for the charge, current and time equation

Worked Example

When will 8 A of current pass through an electrical circuit?

A. When 8 J of energy is used by 1 C of charge

B. When a charge of 4 C passes in 0.5 s

ANSWER: B

- The equation relating current, charge and time is:

$$Q = I \times t$$

- Rearrange to make current *I* the subject of the equation:

$$I = \frac{Q}{t}$$

- Consider option **B**, where Q = 4 C and *t* = 0.5 s:

$$I = \frac{4}{0.5} = 8 \text{ A}$$

- Therefore, the correct answer is **B**

A is incorrect as this is the definition of a voltage of 8 V between two points and does not describe current

C is incorrect as $I = \frac{8}{0.1} = 80 \text{ A}$

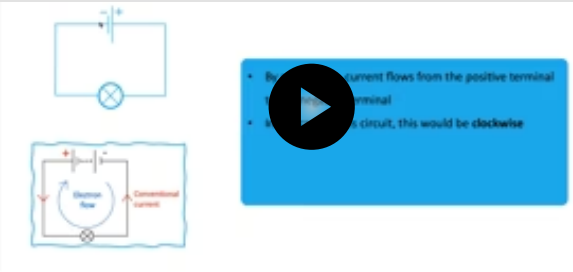
D is incorrect as $I = \frac{1}{8} = 0.125 \text{ A}$

Examiner Tips and Tricks

Electric currents in everyday circuits tend to be quite small, so it's common for examiners to throw in a **unit prefix** like 'm' next to quantities of current, e.g. 10 mA (10 milliamperes). Make sure you can convert these into standard units, e.g. 10 mA = 10 × 10^{-3} A.

Make sure to only use the triangle to help you **rearrange** the equation that links charge, current and time. Don't draw it if you are asked to write out the equation in full, such as Q = *I* × *t*, as you may lose marks for doing so.

Check out this revision note on speed, distance and time if you need a reminder on how to use formula triangles.



Did this video help you? Yes No

Conventional current

Extended tier only

- Current is the flow of **positive** charge i.e. from the **positive** terminal **to** the **negative** terminal of a cell
  - This is known as **conventional current**
- This is in the opposite direction to **electron flow**
  - Electrons are negatively charged, so they flow from the **negative** terminal **to** the **positive** terminal of a cell

Conventional current and electron flow

By definition, conventional current always goes from positive to negative. This is in the opposite direction to the flow of electrons

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